



Public Economics

Lecture 9 · Cost-Benefit Analysis

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- Once we know a public good is underprovided, how does a government decide whether a *specific* project is worth building?
- How do we put a dollar value on things that don't have a market price — saved time, a cleaner river, even a human life?
- How far into the future should we count costs and benefits, and at what rate should we discount them?

Based on

Saez, E., *Econ 131* lecture notes (UC Berkeley); Gruber, J. *Public Finance and Public Policy*, Ch. 8.

8.1 Measuring the Costs of Public Projects

8.2 Measuring the Benefits of Public Projects

8.3 Putting It All Together

Cost-benefit analysis

The comparison of the costs and benefits of a public project, used to decide whether it should be undertaken.

Sounds like simple accounting — add up the costs, add up the benefits, compare. In practice it is genuinely hard, because:

- Many relevant resources are **not** priced in a competitive market.
- Some benefits have **no market price at all**.
- Costs and benefits often arrive at very different **points in time**.

- The government is considering building a 400km train line, connecting two major cities.
- Currently commuters drive from one city to another. It takes about 5 hours by car. The train would only take 2 hours.
- This commuting road is often affected by heavy traffic jams.
- Think of **all the potential benefits** of constructing this train line.

8.1 Measuring the Costs of Public Projects

Cash-flow accounting

Calculates costs by simply adding up what government pays for inputs, and benefits by adding up the income or revenue a project generates.

Opportunity cost

The social marginal cost of any resource is the value of that resource in its **next best use**.

The general rule: economic costs are only those costs that come from **diverting** a resource away from its next best use — not simply what the government happens to pay for it.

Perfectly competitive markets: Social cost = price. True for most labor and material inputs.

Imperfectly competitive markets:

- **Monopoly** (e.g., asphalt supplied by a monopolist): price = marginal cost + monopoly markup > marginal cost. On efficiency grounds, **social cost** = **marginal cost** — the markup is just a transfer from taxpayers to the monopolist, which matters for distribution but not efficiency.
- **Labor market with unemployment:** suppose a \$10 minimum wage creates involuntary unemployment, and the unemployed would work for \$6 on average. A government project hires them at \$10/hour — but the workers gain a \$4/hour surplus. **Social cost** = $\$10 - \$4 = \$6$, not \$10.

Present discounted value (PDV)

A dollar next year is worth $1/(1+r)$ of a dollar today, since that dollar could otherwise earn interest r .

A maintenance cost of F paid forever has PDV:

$$\frac{F}{1+r} + \frac{F}{(1+r)^2} + \frac{F}{(1+r)^3} + \cdots = \frac{F}{r}$$

Government typically discounts at its own borrowing rate r (e.g., 6% nominal, 3% real). This is reasonable for **short-run** projects — but becomes problematic over the long run: at $r = 3\%$, \$1 in 100 years is worth only \$0.052 today. Long-run costs like climate change get discounted almost to nothing.

Social discount rate

The appropriate rate r for computing the PDV of a *social* investment.

Two distinct reasons to discount a future dollar relative to a dollar today:

1. **Absolute (“pure time”) discounting:** people simply prefer \$1 now to \$1 later. On ethical grounds, it’s not obvious we should discount *future generations’* welfare this way — except for genuine existential risk (e.g., a meteorite ending civilization).
2. **Economic growth:** future generations will likely be richer, so an extra dollar is worth less to them at the margin — this alone justifies some discounting even with zero pure time preference.

In an ideal world, both effects are already embedded in the market interest rate r , so using today’s r would be enough.

The problem: we don't actually know how growth — and hence r — will evolve over the next century. If the economy collapses (e.g., from unchecked climate change), future generations will be **poor**, and we should *not* discount their welfare heavily.

A simple illustration

50% chance of zero growth ($r = 0\%$): \$1 in 100 years = \$1 today.

50% chance of normal growth ($r = 3\%$): \$1 in 100 years = \$0.052 today.

Expected value today: $0.5(1) + 0.5(0.052) = \$0.552$.

Solving $(1 + \bar{r})^{-100} = 0.552$ gives an implied discount rate $\bar{r} \approx 0.6\%$ — far below either scenario's own rate.

Takeaway

When there's a real chance growth stops, the distant future should be discounted at a **low** rate.

8.2 Measuring the Benefits of Public Projects

If individuals freely optimize their labor supply, they're indifferent between working one more hour and enjoying one more hour of leisure at the margin.

Hourly wage = value of one hour of leisure

So a project that saves people time (e.g., faster commutes) can be valued using wages — whether people use the saved time to work more or relax more. **Problems in**

practice:

1. Individuals often can't freely trade off hours and leisure — many jobs come with fixed-hours constraints.
2. An hour *stuck in traffic* is worse than an hour of ordinary leisure lost — so the true value of reducing congestion exceeds the wage-based estimate.

Contingent valuation

Directly asking individuals to value an option they are not currently choosing, or cannot choose at all.

Often the **only** feasible method when there is no market price whatsoever:

- The value of saving an endangered species.
- The value of keeping the Arctic pristine.

Popular among environmental advocates as a way to argue that a cause is “worth” a large number — which is exactly where the method’s problems begin.

Diamond & Hausman show that the structure of a contingent-valuation survey can swing the answer wildly:

- **No real cost to respondents** — easy to exaggerate value when you don't actually have to pay.
- **Isolation effects:** asking about one cause alone vs. many causes together changes the stated value.
- **Order effects:** which question comes first matters.
- **Embedding effect:** people state a similar willingness to pay for saving *one* lake as for saving *three* lakes.

The right comparison

A rational allocation requires looking at *all* competing causes together — weighing one cause in isolation is not a sound basis for public policy. Government, which must set a single budget across all of them, is best placed to make that trade-off.

Revealed preference (hedonic approach)

Let people's actual choices reveal their valuation.

- **1970s US gas price controls** created queues at most stations — but small independent stations, exempt from the controls, could charge more and had shorter queues. Comparing the price gap to the queue-length gap: saving \$10 by queuing one hour $\Rightarrow$ one hour is worth about \$10.
- **FasTrak toll lanes** (San Francisco Bay Area): what drivers pay to skip congestion reveals the value of time *and* of avoiding the unpleasantness of queuing itself.

These estimates capture the value of time for the *marginal* person — the one just indifferent between paying and waiting — not necessarily the average person.

Valuing human life is the single hardest, most ethically fraught issue in cost-benefit analysis.

- Yet virtually *every* government expenditure changes the odds of saving a life somewhere — safer roads, health care spending, and so on.
- So some value must be placed on a **statistical** life — a small change in the probability of death across a large population.

A crucial distinction

We can meaningfully price a **statistical** life (fewer expected accidents across a population) even though pricing a specific, identified, **real** life is neither possible nor something anyone would attempt.

As with time, economists' preferred method is revealed preference: how much are people actually willing to pay to reduce their own odds of dying?

Compensating differentials

Extra (or reduced) wages paid to compensate workers for a job's negative (or positive) amenities — including mortality risk.

Example: a \$10,000 enlistment bonus was needed to recruit soldiers during the Afghanistan/Iraq wars, which carried an extra 1/1,000 chance of dying relative to peacetime.

$$\text{Value of a statistical life} = \frac{\$10,000}{1/1,000} = \$10 \text{ million}$$

Limitations: requires rational behavior, and again values the *marginal* person's life, not the average.

Source: US studies put the revealed value of a statistical life at about \$9.3 million on average today.

Ashenfelter & Greenstone (2004), *JPE*: in 1987 the US federal government allowed states to raise rural highway speed limits from 55 to 65 mph — 21 states did so.

- The higher limit raised speeds by $\approx 3.5\%$ and raised fatality rates by $\approx 35\%$.
- That trade-off implies about **125,000 hours** of travel time saved per additional life lost.
- Valuing time saved at the average hourly wage: adopting states were effectively willing to accept a fatality risk in exchange for saving **\$1.54 million** (1997 dollars) per statistical life.

Interpretation

These states were revealing a value of a statistical life of **at most** \$1.54 million — well below the \$9–10 million found in other contexts, a useful caution about how much these estimates vary by setting.

8.3 Putting It All Together

- **Counting secondary benefits:** a new highway boosts commerce nearby — but often just *relocates* activity from somewhere else rather than creating it.
- **Counting labor as a benefit:** jobs “created” by a project are a **cost**, not a benefit — those workers are not producing something else while employed on this project.
- **Double-counting benefits:** a project that reduces commuting time *and* raises nearby house values is often counting the **same** benefit twice, since the house-price gain largely capitalizes the time savings.

Distributional concerns

The costs and benefits of a public project rarely fall on the same people. A project can pass a cost-benefit test in aggregate while still making some groups clearly worse off.

Uncertainty

Both the costs and the benefits of public projects are often highly uncertain — construction costs run over budget, and benefits (time saved, lives saved, growth effects) depend on assumptions that may not hold.

A cost-benefit number is a starting point for a policy debate, not a verdict that ends it.

Thank you!

Questions and discussion